Individual Documentation Of Disease Detection_AI-Chatbot

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1) Problem Description

Accurate disease prediction based on patient symptoms is a challenging task in the healthcare sector. Many diseases share common symptoms, making manual diagnosis difficult and time-consuming. Early disease detection can assist patients in obtaining timely medical consultation and treatment.

The objective of this project is to develop an intelligent Disease Detection AI Chatbot capable of predicting diseases from user-provided symptoms. The system uses machine learning techniques to learn symptom-disease relationships from historical medical datasets. After training, the model can identify the most probable disease and provide relevant information and precautionary measures.

The chatbot serves as a preliminary diagnostic support system and helps bridge the gap between symptom appearance and professional medical consultation.

2) Methodology

Our project follows a Supervised Machine Learning approach.

● Model Selection:

XGBoost (Extreme Gradient Boosting) was selected as the primary classification model. XGBoost is one of the most powerful ensemble learning algorithms and is widely used because of its high predictive performance, regularization capability, and efficient handling of multiclass classification problems.

● **Data Preparation:**

The dataset was loaded from CSV files containing symptom features and disease labels. Duplicate columns were removed, and column names were standardized to ensure consistency during training and prediction.

● **Evaluation Strategy:**

The dataset was divided into training and testing subsets using a 67%-33% split. Model performance was evaluated using Accuracy, Precision, Recall, and F1-Score. In addition, 5-Fold Cross Validation was applied to measure the model's stability and robustness across multiple data partitions.

3) Implementation Details

● **Data Preprocessing:**

- Dataset loaded using Pandas.
- Duplicate symptom columns removed.

- Disease labels encoded using LabelEncoder.
- Features and target labels separated.
- Data split into training and testing sets using train_test_split().

● **Symptom Processing:**

The chatbot includes a symptom extraction mechanism that detects symptoms from natural language user input. Synonym mapping and fuzzy matching techniques are used to recognize symptoms even when users provide alternative spellings or phrases.

Examples:

stomach ache → stomach_pain

loose motion → diarrhea

throat pain → sore_throat

body ache → muscle_pain

● **Backend & Model:**

The backend was implemented in Python. The XGBoost classifier was trained using symptom vectors as input features and disease names as output labels. After training, the model predicts the most probable disease along with confidence scores generated from prediction probabilities.

The chatbot also retrieves:

- Disease descriptions
- Symptom severity information
- Recommended precautions
- from external knowledge files.

4) Results & Analysis

Model Performance:

Training Accuracy : 96.69%

Testing Accuracy : 85.25%

CV Accuracy : 92.74%

Precision : 83.77%

Recall : 85.25%

F1-score : 83.90%

Analysis:

The XGBoost model achieved a training accuracy of 96.69%, indicating that the model learned the training data very well. However, the testing accuracy of 85.25% shows a noticeable drop compared to training performance, suggesting a moderate level of overfitting.

The cross-validation accuracy of 92.74% indicates that the model maintains relatively strong and stable performance across different data splits, although some variation exists between folds.

The precision of 83.77% suggests that when the model predicts a disease, it is correct most of the time, but there are still some false positives. The recall of 85.25% indicates that the model is able to correctly identify most actual disease cases, though a few cases are missed. The F1-score of 83.90% provides a balanced view of precision and recall, confirming overall consistent but not perfect performance.

The confusion matrix analysis shows that while most disease classes are correctly classified, some misclassifications occur between diseases with similar symptom patterns. This is expected in symptom-based prediction systems, as multiple diseases can share overlapping symptoms.

Overall, the XGBoost-based Disease Detection AI Chatbot demonstrates strong predictive capability and reasonable generalization performance. However, further improvement can be achieved by increasing dataset diversity, feature engineering, and incorporating real-world clinical data.